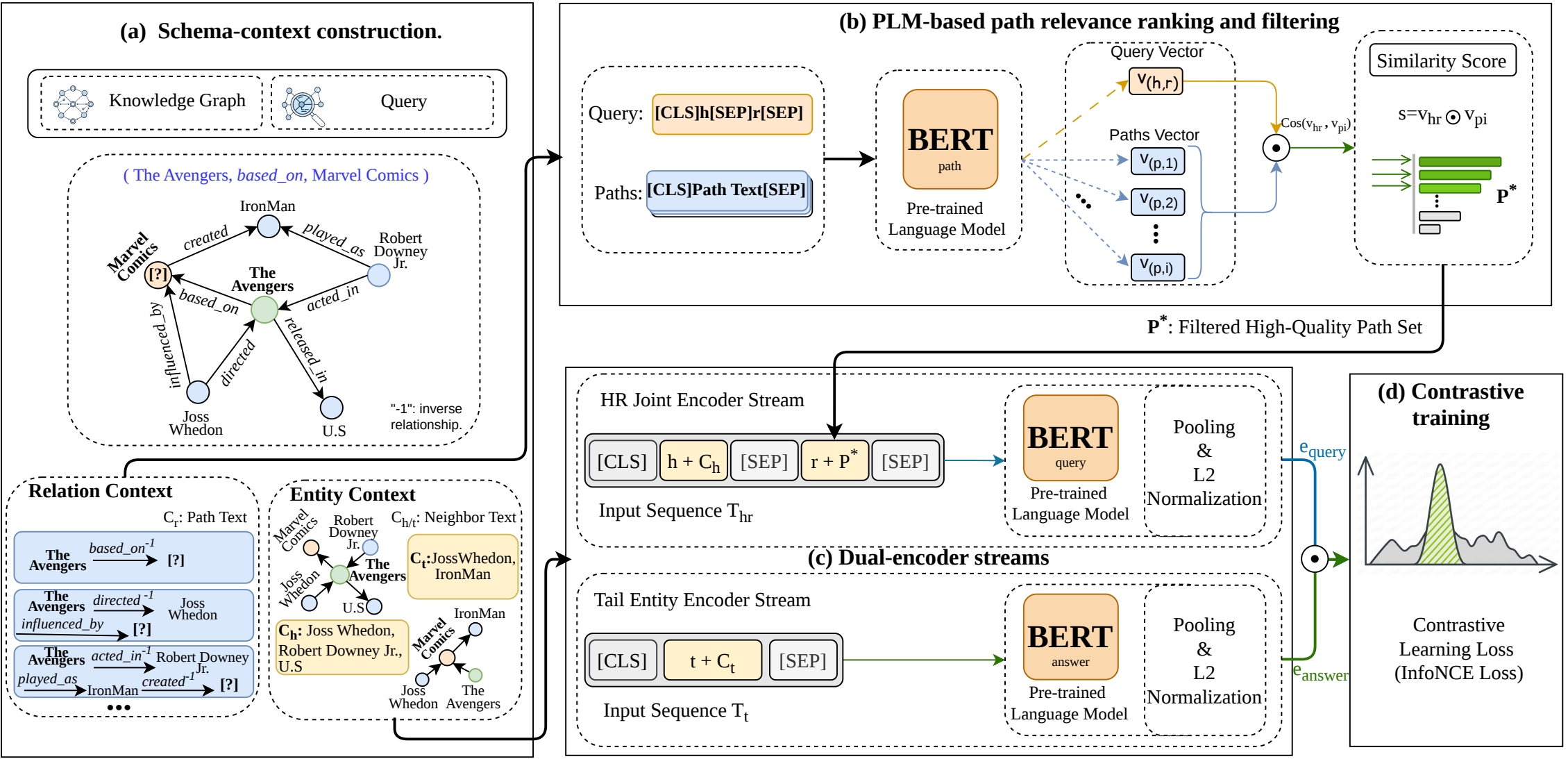

(a) Schema-context construction.
Knowledge Graph
Query
(The Avengers, based_on, Marvel Comics)
IronMan
Marvel Comics
created
played_as
Robert Downey Jr.
[?]
The Avengers
based_on
acted_in
influenced_by
directed
released_in
Joss Whedon
U.S
"-1": inverse relationship.
Relation Context
C_r: Path Text
The Avengers based_on^{-1} [?]
The Avengers directed^{-1} Joss Whedon influenced_by [?]
The Avengers acted_in^{-1} Robert Downey Jr. played_as IronMan created^{-1} [?]
Entity Context
$C_{h/t}$: Neighbor Text
C_t:JossWhedon, IronMan
C_h: Joss Whedon, Robert Downey Jr., U.S
(b) PLM-based path relevance ranking and filtering
Query: [CLS]h[SEP]r[SEP]
Paths: [CLS]Path Text[SEP]
BERT path
Pre-trained Language Model
Query Vector
$v_{(h,r)}$
Paths Vector
$v_{(p,1)}$
$v_{(p,2)}$
$v_{(p,i)}$
$\cos(v_{hr}, v_{pi})$
Similarity Score
$s=v_{hr} \odot v_{pi}$
P^*
P^*: Filtered High-Quality Path Set
HR Joint Encoder Stream
[CLS] h + C_h [SEP] r + P^* [SEP]
Input Sequence T_{hr}
BERT query
Pre-trained Language Model
Pooling & L2 Normalization
(c) Dual-encoder streams
Tail Entity Encoder Stream
[CLS] t + C_t [SEP]
Input Sequence T_t
BERT answer
Pre-trained Language Model
Pooling & L2 Normalization
e_{query}
e_{answer}
(d) Contrastive training
Contrastive Learning Loss (InfoNCE Loss)